POSTS AND TELECOMMUNICATIONS
INSTITUE OF TECHNOLOGY
INFORMATION TECHNOLOGY

INTERNSHIP

REPORT

WoxionChat

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1 Overview

Building upon the robust backend foundation established last week, the focus of this week shifted towards the AI core of **WoxionChat**. We successfully developed the automated data pipeline and generated a comprehensive dataset required for fine-tuning our domain-specific Large Language Model (LLM).

2 Technical Accomplishments

2.1 AI Data Pipeline Development

A scalable and automated pipeline was implemented to prepare raw data for model training:

[label=-]

- **Data Ingestion & Cleaning:** Developed scripts to aggregate chat logs and documentation, followed by rigorous cleaning to remove PII (Personally Identifiable Information) and noise.
- **Automated Preprocessing:** Implemented tokenization, normalization, and deduplication logic to ensure high-quality input for the training phase.
- **Validation Layer:** Integrated a validation step to check for data consistency and structural integrity within the NoSQL storage.

2.2 Finetuning Dataset Generation

We successfully generated a specialized dataset tailored for the WoxionChat ecosystem:

- **Synthetic Data Generation:** Leveraged advanced prompting techniques to generate high-quality instruction-response pairs covering common user queries and technical support scenarios.
- **Diversity & Balance:** Ensured the dataset contains a balanced distribution of intent categories, including general conversation, technical troubleshooting, and administrative commands.
- **Formatting:** All generated data has been exported into JSONL format, optimized for training frameworks such as Hugging Face or PyTorch.

2.3 AI Workflow Orchestration

- **Flow Design:** Completed the conceptual and logic design for the AI response cycle, integrating context management to ensure coherent long-form conversations.
- **System Integration:** Prepared the Django backend hooks to consume AI model outputs via asynchronous task queues (Celery/Redis).

3 Technology Stack Update

- **Data Tools:** Python (Pandas, NumPy), JSONL format.
- **AI Models:** Utilized Base LLMs for synthetic data augmentation.
- **Storage:** MongoDB (storing raw and processed training samples).

4 Progress Evaluation

- **Status:** On Track.
- **Observations:** The quality of the generated synthetic data is high, which is expected to significantly reduce hallucination rates in the production model. The pipeline is now ready for iterative data updates.

5 Upcoming Milestone (Next Week)

The next phase will focus on the actual training and initial testing:

1. **Model Training Execution:** Initiating the fine-tuning process using the generated dataset on high-performance compute nodes.
2. **Evaluation & Benchmarking:** Measuring model performance against baseline metrics (latency, accuracy, and tone consistency).
3. **Admin Interface Integration:** Finalizing the UI for "The Boss" to inspect AI performance and manage the fine-tuning logs.